

Design and Analysis of Randomized Trials in Public Health

1. You have developed two interventions to reduce fasting blood glucose in an adult population at risk for diabetes. One intervention is dietary (“diet”) and compares two diets (“diet A” or “diet B”) to people’s typical diets (“standard diet”). The other intervention is behavioral (“behavior”), in which the spouse or partner of a participant is also engaged in the dietary intervention (“partner engagement”) or not (“no partner engagement”). You evaluate these interventions in a 3 x 2 factorial design with 180 participating households in total. At the conclusion of your interventions, you measure fasting blood glucose (and other variables) in your enrolled participants.
 - a) Write the statistical model for an intention-to-treat analysis in which you consider the effect of the dietary intervention, the behavioral intervention, and their interaction on fasting blood glucose. Define all notation you use and all assumptions your model makes. (3 points)
 - b) Your intention-to-treat analysis yields the overall ANOVA table below. Fill in the empty cells. (3 points)

Source	DF	Sum of Squares	Mean Square	F Value
Model				
Error		20147.66		
Corrected Total		36344.20		

- c) Your type III tests are below. Fill in the empty cells. (3 points)

Source	DF	Type III SS	Mean Square	F Value	Pr > F
diet		11321.47			<.0001
behavior		3738.27			<.0001
diet*behavior		185.94			0.4497

- d) In the table in part (c), the p-value for “diet” corresponds to a test of which null hypothesis? Write the null hypothesis using both lay terms and the notation you used in part (a). (2 points)
 - e) When testing the hypothesis in part (d), the F value you calculated is compared to the critical value from what distribution? Make sure to specify degrees of freedom, if necessary. (2 points)
 - f) Which will be greater: the F value you calculated or that critical value? (1 point)

- g) Does the behavioral intervention have a consistent effect on blood glucose or is its effect modified by the dietary intervention? How do you know? (1 point)

Design and Analysis of Randomized Trials in Public Health

1. You have developed two interventions to reduce fasting blood glucose in an adult population at risk for diabetes. One intervention is dietary (“diet”) and compares two diets (“diet A” or “diet B”) to people’s typical diets (“control”). The other intervention is behavioral (“behavior”), in which the spouse or partner of a participant is also engaged in the dietary intervention (“spouse”) or not (“control”). You evaluate these interventions in a 3 x 2 factorial design with 180 participating households in total. At the conclusion of your interventions, you measure fasting blood glucose (and other variables) in your enrolled participants.

- a) Write the statistical model for an intention-to-treat analysis in which you consider the effect of the dietary intervention, the behavioral intervention, and their interaction on fasting blood glucose. Define all notation you use and all assumptions your model makes. (3 points)

$$Y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}$$

Y_{ij} is the fasting blood glucose of the k th participant assigned to the i th diet and the j th behavioral intervention

μ is the average fasting blood glucose of the population

τ_i is the effect of the i th diet on a person’s fasting blood glucose

β_j is the effect of the j th behavioral intervention on a person’s fasting blood glucose

$(\tau\beta)_{ij}$ is the interaction effect for the i th diet and j th behavioral intervention

ε_{ijk} is the random error term for the k th participant assigned to the i th diet and the j th behavioral intervention

$i = 1, 2, 3$

$j = 1, 2$

$\varepsilon_{ijk} \sim \text{iid } N(0, \sigma^2)$

- b) Your intention-to-treat analysis yields the overall ANOVA table below. Fill in the empty cells. (3 points)

Source	DF	Sum of Squares	Mean Square	F Value
Model	5	16196.54	3239.31	27.98
Error	174	20147.66	115.79	
Corrected Total	179	36344.20		

- c) Your type III tests are below. Fill in the empty cells. (3 points)

Source	DF	Type III SS	Mean Square	F Value	Pr > F
diet	2	11321.47	5660.74	48.89	<.0001
behavior	1	3738.27	3738.27	32.28	<.0001
diet*behavior	2	185.94	92.97	0.80	0.4497

Note the F statistics are calculating by dividing the respective mean square by the mean squared error calculated in part (b).

- d) In the table in part (c), the p-value for “diet” corresponds to a test of which null hypothesis? Write the null hypothesis using both lay terms and the notation you used in part (a). (2 points)

The null hypothesis is the average fasting blood glucose level in this population is the same regardless of which of the three diets is being consumed, i.e., there is no difference in average fasting blood glucose among people who consume these three diets.

$$H_0 : \mu_{\text{diet A}} = \mu_{\text{diet B}} = \mu_{\text{standard}} \quad \text{or} \quad H_0 : \mu_1 = \mu_2 = \mu_3$$

- e) When testing the hypothesis in part (d), the F value you calculated is compared to the critical value from what distribution? Make sure to specify degrees of freedom, if necessary. (2 points)

$$F_{2,174}$$

- f) Which will be greater: the F value you calculated or that critical value? (1 point)

Assuming we used a significance level of $\alpha=0.05$, the F value we calculated (48.89) will greater than the critical value because $p < \alpha$

- g) Does the behavioral intervention have a consistent effect on blood glucose or is its effect modified by the dietary intervention? How do you know? (1 point)

The behavioral intervention has a consistent effect on blood glucose: the main effect is significant ($p < 0.0001$) while the interaction with diet is not ($p = 0.4497$). If the interaction between the behavioral intervention and the dietary intervention were significant, we would conclude that the effect of behavioral intervention was modified by the dietary intervention (or vice versa).